

Podcast 14 — Auto Scaling & Load Balancing

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Length target: 15 min **Repo references:** `domain-2-new-solutions/33-ec2-auto-scaling.md` , `29-elb.md` , `domain-3-continuous-improvement/08-application-auto-scaling.md`

Topic & Scope

EC2 Auto Scaling Groups + Application Auto Scaling + Elastic Load Balancing. SAP-C02 tests which scaling service applies to which resource, and the four ELB types.

Service Coverage Depth

Deep: - EC2 Auto Scaling — launch templates, scaling policies, lifecycle hooks, instance refresh - Application Auto Scaling — DynamoDB, Aurora replicas, ECS service, Lambda PC, etc. - ELB types: ALB, NLB, GWLB, Classic - Target groups, health checks, deregistration delay

Brief: - AWS Auto Scaling (unified scaling plans console) - Predictive scaling

Structured Outline

1. **Open (30s)** — “Three different ‘Auto Scaling’ services sound the same. Plus four load balancer types. Untangle them now.”
2. **EC2 Auto Scaling (4 min)** — ASG, launch templates, three scaling types (target tracking, step, simple), lifecycle hooks, instance refresh, mixed instance policies, warm pools
3. **Application Auto Scaling (3 min)** — for DynamoDB / Aurora replicas / ECS / Lambda PC / SageMaker / Spot Fleet / AppStream / Comprehend / Neptune / ElastiCache / Keyspaces
4. **ELB Types (4 min)** — ALB (HTTP/HTTPS, content routing, WAF-compatible), NLB (TCP/UDP, static IP, ultra-low-latency), GWLB (security appliance insertion), Classic (legacy)
5. **Target Groups + Health Checks (1.5 min)** — multiple TGs per ALB, health check types, deregistration delay (connection draining), cross-zone load balancing
6. **AWS Auto Scaling + Predictive (1 min)** — unified console, ML-based forecast
7. **Rapid-Fire Trap Drill (60s)**

Must-Mention Exam Tips

- EC2 Auto Scaling: ASG with launch template + min/max/desired
- Scaling policies: target tracking (simplest), step scaling, simple scaling, scheduled scaling
- Lifecycle hooks: pause launch/terminate to run setup/cleanup
- Instance refresh: rolling replacement after launch template change
- Mixed instances policy: combine On-Demand + Spot in one ASG

- Warm pools: pre-initialized instances to reduce scale-out latency
- Application Auto Scaling supports: DynamoDB, Aurora Replicas, ECS, Lambda Provisioned Concurrency, SageMaker, Spot Fleet, AppStream, Comprehend, Neptune, ElastiCache, Keyspaces, EMR, Custom resources
- ALB: HTTP/HTTPS, path/host/header routing, WebSockets, can route to EC2/IP/Lambda/containers
- NLB: TCP/UDP/TLS, static IPs per AZ, ultra-low-latency, preserves source IP
- GWLB: insert third-party network appliances (firewalls, IDS), uses GENEVE protocol
- Classic Load Balancer: legacy, avoid for new
- Target groups: separate per protocol/port, supports lambda targets (ALB only)
- Health checks: TCP / HTTP / HTTPS with path; configurable thresholds
- Deregistration delay: connection draining window (default 300s)
- Cross-zone load balancing: ALB always on (free); NLB off by default (paid when on)
- Predictive scaling: ML-based forecast for cyclical workloads, available for EC2 ASG and AWS Auto Scaling plans

Must-Mention Exam Traps

- EXAM TRAP: EC2 Auto Scaling $\neq$ Application Auto Scaling — first is EC2 ASGs; second is for non-EC2 resources
- EXAM TRAP: AWS Auto Scaling is the unified CONSOLE — most use per-service Application Auto Scaling directly
- EXAM TRAP: DynamoDB on-demand mode auto-scales internally — Application Auto Scaling only configures provisioned mode
- EXAM TRAP: ALB does NOT preserve source IP by default — uses X-Forwarded-For header; NLB preserves natively
- EXAM TRAP: ALB cannot do static IP — use Global Accelerator + ALB, or use NLB
- EXAM TRAP: NLB cross-zone is OFF by default — enabling costs extra
- EXAM TRAP: WAF attaches to ALB but NOT NLB
- EXAM TRAP: GWLB inserts third-party appliances — not by itself a firewall
- EXAM TRAP: Aurora Auto Scaling scales REPLICAS, not the writer — for writer scaling, use Aurora Serverless v2
- EXAM TRAP: ASG health checks include EC2 status + ELB health — must enable ELB health checks explicitly for ELB-based replacement
- EXAM TRAP: Spot in ASG via mixed instances policy — pure Spot ASG is risky for prod
- EXAM TRAP: Compute Optimizer recommends ASG sizing — don't confuse with Auto Scaling itself

Key Decision Matrix

Scenario	Pick
Scale EC2 instances by CPU / requests	EC2 Auto Scaling target tracking
Scale DynamoDB capacity	Application Auto Scaling
Scale Aurora read replicas	Application Auto Scaling
Scale Lambda Provisioned Concurrency	Application Auto Scaling
HTTP load balancing with path-based routing	Application Load Balancer
TCP/UDP ultra-low-latency LB	Network Load Balancer
Static IPs for whitelisting	NLB (or Global Accelerator + ALB)
Insert firewall appliance inline	Gateway Load Balancer
Lambda as target	ALB
Preserve source IP natively	NLB

Tone & Style

- Three “Auto Scaling” framing — emphasize they are different
- Repeat ALB = HTTP, NLB = TCP/UDP throughout
- Use concrete examples: “Lambda PC scaling? Application Auto Scaling”

Rapid-Fire Closer

“Scale DynamoDB?” — “Application Auto Scaling.” “Static IP LB?” — “NLB.” “Path-based routing?” — “ALB.” “Insert third-party firewall?” — “GWLB.” “Aurora writer scale?” — “Serverless v2, NOT replicas.”